Two-Camera IR Mocap Hover

Frames

$$
\begin{aligned}
[\cdot]^W &: \text{world} \quad (\text{ArUco marker center}, +x, +y \text{ marker edges}, +z \text{ up}) \\
[\cdot]^B &: \text{body} \quad (+x \text{ nose}, +y \text{ portside}, +z \text{ up}) \\
[\cdot]^{C_i} &: \text{camera } i \quad (+x \text{ right}, +y \text{ down}, +z \text{ optical axis})
\end{aligned}
\tag{1}
$$

$[T]^{YX}$: rotation coordinatizing an X-frame tensor in Y, $[a]^Y = [T]^{YX}[a]^X$
$[T]^{XY} = \left([T]^{YX}\right)^\top$ $[a]_\times$: skew matrix, $[a]_\times b = a \times b$
$\mathrm{Exp}(\phi)$: rotation of angle $\|\phi\|$ about $\phi/\|\phi\|$ (Rodrigues)
$\pi(\cdot)$: perspective divide, $\pi\left(\begin{pmatrix} a \\ b \\ c \end{pmatrix}\right) = \begin{pmatrix} a/c \\ b/c \end{pmatrix}$

Camera pose

Problem

One flat marker fixes W; each camera solves its own pose against it.

$$
\left[X_j\right]^W = (\pm L/2, \pm L/2, 0)^\top, \quad j = 1 \ldots 4 \tag{2}
$$

$$
s \begin{pmatrix} u \\ v \\ 1 \end{pmatrix}_j = K\left([T]^{CW}\left[X_j\right]^W + [t]^C\right) \tag{3}
$$

$\left[X_j\right]^W$: marker corners [m] L: marker edge [m]
u_j, v_j: measured corner pixels [px] K: intrinsics
$[T]^{CW}, [t]^C$: pose of W as seen from C

Solution

$$
\left[\hat{T}\right]^{CW}, \left[\hat{t}\right]^C = \arg\min_{T,t} \sum_{j=1}^{4} \left\| \pi\left(K\left([T]^{CW}\left[X_j\right]^W + [t]^C\right)\right) - \begin{pmatrix} u \\ v \end{pmatrix}_j \right\|^2 \tag{4}
$$

Closed form for a planar square (IPPE), which returns both mirror poses; keep the smaller residual, report the ratio as an ambiguity check.

$$
[c]^W = -[T]^{WC}[t]^C \tag{5}
$$

$[c]^W$: camera center [m]

Drone pose

Triangulation

Per LED, from $x \times PX = 0$, two rows per camera.

$$
A[X]^W = 0, \quad A = \begin{pmatrix} u_0 p_0^{3\top} - p_0^{1\top} \\ v_0 p_0^{3\top} - p_0^{2\top} \\ u_1 p_1^{3\top} - p_1^{1\top} \\ v_1 p_1^{3\top} - p_1^{2\top} \end{pmatrix} \tag{6}
$$

$P_i = K\left[[T]^{C_i W} \mid [t]^{C_i}\right]$: projection matrix $p_i^{k\top}$: its row k
u_i, v_i: undistorted pixels of one LED in camera i [px]
$[X]^W$: right singular vector of A at $\sigma_{\min}$, dehomogenized [m]

Correspondence

Blobs arrive unlabeled; $3! = 6$ pairings, take the smallest mean reprojection error.

$$\sigma^* = \underset{\sigma \in S_3}{\arg\min}\ \frac{1}{3} \sum_{k=1}^{3} \sum_{i=0}^{1} \left\| \pi\left(P_i \left[X_{k,\sigma}\right]^W\right) - \begin{pmatrix} u \\ v \end{pmatrix}_{i,k} \right\|^2 \tag{7}$$

Rigid fit (Kabsch)

$$H = \sum_{k=1}^{3} \left([b_k]^B - [\bar{b}]^B\right)\left([w_k]^W - [\bar{w}]^W\right)^\top = U\Sigma V^\top \tag{8}$$

$$[T]^{WB} = V \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(VU^\top) \end{pmatrix} U^\top, \quad [p]^W = [\bar{w}]^W - [T]^{WB} [\bar{b}]^B \tag{9}$$

$$\mathrm{rmsd} = \sqrt{\frac{1}{3} \sum_{k=1}^{3} \left\| [T]^{WB} [b_k]^B + [p]^W - [w_k]^W \right\|^2} \tag{10}$$

$[b_k]^B$: LED positions on the airframe [m] $[w_k]^W$: triangulated LEDs [m]
$[T]^{WB}, [p]^W$: drone attitude and position
$\det(VU^\top)$ term: forces a rotation, not a reflection
rmsd: fit residual [m], also the quality gate
labeling of $[w_k]^W$ against $[b_k]^B$: chosen by the same minimization

LQR control law

Problem

One axis of $[\cdot]^W$, error $e = z - z_{\mathrm{ref}}$. Betaflight holds attitude; the commanded lean arrives through a first-order lag.

$$x = \begin{pmatrix} \int e \\ e \\ \dot{e} \\ a \end{pmatrix}, \quad u = a_{\mathrm{cmd}}, \quad \dot{x} = Ax + Bu \tag{11}$$

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1/\tau \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1/\tau \end{pmatrix} \tag{12}$$

$$J = \sum_{k=0}^{\infty} x_k^\top Q x_k + u_k^\top R u_k \tag{13}$$

a: acceleration actually delivered, chasing u [m/s^2]
τ: attitude (or throttle) lag + transport delay [s]
$Q = \mathrm{diag}\left(1/x_{i,\max}^2\right)$, $R = \rho^2/a_{\max}^2$: Bryson weights
ρ: `effort`, trades aggression against smoothness

Solution

Zero-order hold at the loop rate, then the discrete Riccati equation.

$$\begin{pmatrix} A_d & B_d \\ 0 & I \end{pmatrix} = \exp\left(\begin{pmatrix} A & B \\ 0 & 0 \end{pmatrix} \Delta t\right) \tag{14}$$

$$P = A_d^\top P A_d - A_d^\top P B_d \left(R + B_d^\top P B_d\right)^{-1} B_d^\top P A_d + Q \tag{15}$$

$$K = \left(R + B_d^\top P B_d\right)^{-1} B_d^\top P A_d, \quad u = -Kx \tag{16}$$

P: solved by iterating the recursion to convergence (no SciPy)

Command mapping

Coordinatize the demand in B using the estimated yaw ψ, then map to sticks.

$$[a]^B = \begin{pmatrix} a_{\text{fwd}} \\ a_{\text{left}} \\ a_z \end{pmatrix} = [T]^{BW}[a]^W, \quad [T]^{BW} = \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{17}$$

$$\theta = \arctan(a_{\text{fwd}}/g), \quad \varphi = \arctan(-a_{\text{left}}/g), \quad \text{thr} = \text{ff}\,(1 + a_z/g)/\cos\theta \tag{18}$$

θ, φ: pitch, roll commands [rad], clamped to θ_{max}
ff: hover feed-forward, learned in flight from the standing vertical integral
yaw: left to the flight controller

EKF

State

Nominal kept as is; the filter propagates the error state, so the rotation needs no global parameterization.

$$\delta x = \begin{pmatrix} [\delta p]^W \\ [\delta v]^W \\ [\delta\theta]^B \\ [\delta\omega]^B \end{pmatrix} \in \mathbb{R}^{12}, \quad [T]^{WB}_{\text{true}} = [T]^{WB}\,\text{Exp}([\delta\theta]^B) \tag{19}$$

$[p]^W, [v]^W$: position, velocity [m], [m/s]
$[T]^{WB}, [\omega]^B$: attitude, body rate [rad/s]
error on the right: acts in B, keeps H simple, no Euler singularity

Prediction

Constant velocity, constant body rate; acceleration is process noise.

$$[p]^W \leftarrow [p]^W + [v]^W \Delta t, \quad [T]^{WB} \leftarrow [T]^{WB}\,\text{Exp}([\omega]^B \Delta t), \quad P \leftarrow FPF^\top + Q \tag{20}$$

$$F = \begin{pmatrix} I & I\Delta t & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & \text{Exp}([\omega]^B \Delta t)^\top & I\Delta t \\ 0 & 0 & 0 & I \end{pmatrix} \tag{21}$$

Measurement model

Three LEDs, $z \in \mathbb{R}^9$.

$$h_k = [T]^{WB}[b_k]^B + [p]^W, \quad H_k = \left[I \;\; 0 \;\; -[T]^{WB}\left[[b_k]^B\right]_\times \;\; 0 \right] \tag{22}$$

z: triangulated $[w_k]^W$ [m], labeled against the **prediction**
$R_m = \sigma_m^2 I_9$: triangulation noise

Update

Gate first, then Joseph form.

$$y = z - h, \quad S = HP^-H^\top + R_m, \quad y^\top S^{-1} y \le \chi^2_{\text{gate}} \tag{23}$$

$$K = P^-H^\top S^{-1}, \quad \delta x = Ky, \quad P^+ \leftarrow (I - KH)P^-(I - KH)^\top + KR_mK^\top \tag{24}$$

$$[p]^{W+} = [p]^{W-} + [\delta p]^W, \quad [T]^{WB+} = [T]^{WB-}\,\text{Exp}([\delta\theta]^B) \tag{25}$$

y: innovation S: innovation covariance K: Kalman gain
gate rejected, or no measurement: predict only, invalid past t_{coast}
$[T]^{WB+}$: re-orthonormalized by SVD